MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UN
Dept. of Information and Communication Technol
Class Test No. 05
4th Year 1st Semester, Session 2021-2022
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ID No. IT22017 Date
Course Code ICT 4102 Course Title Digital Image Processing

20

Ans. to the ques. no- 01

Compression ratio: Compression ratio is the ratio of the original image size to the compressed image size.

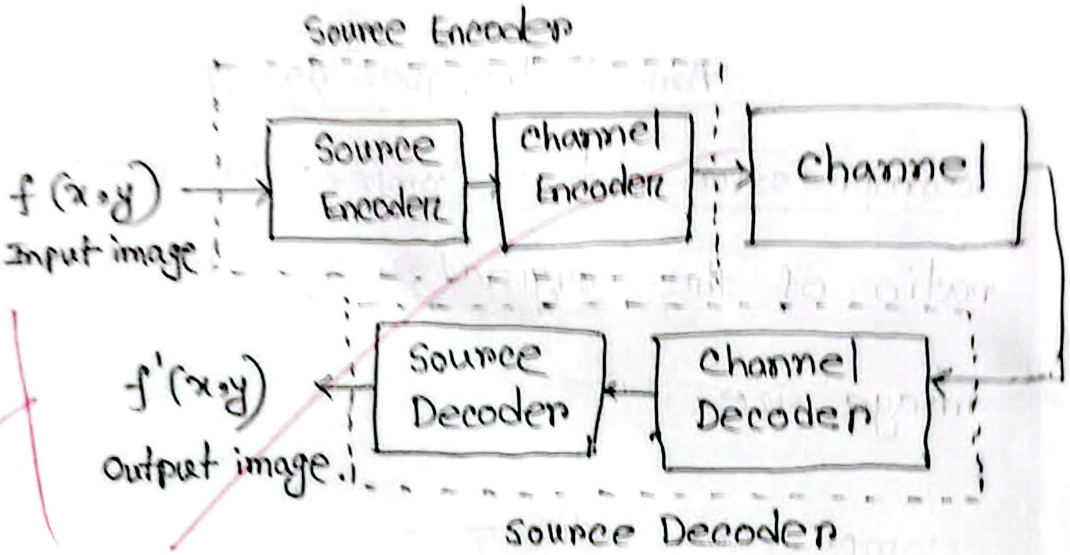
$$\text{Compression ratio} = \frac{\text{Original Image size}}{\text{Compressed image size}}$$

$$\therefore \text{Compression ratio, } C = \frac{n_1}{n_2}$$

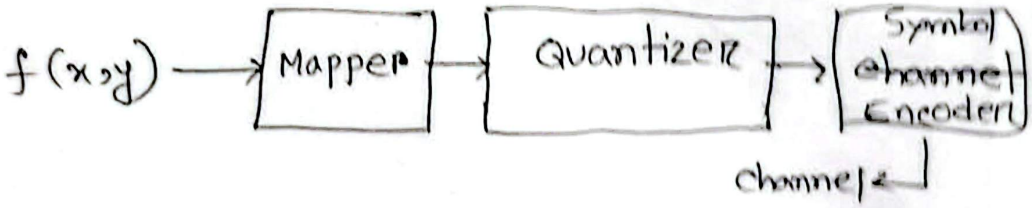
P.T.O

CamScanner

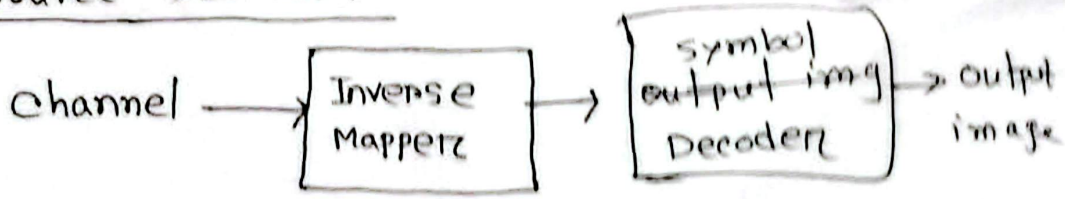
Major building blocks of an image compression model.



Source Encoder :



Source Decoder :



Ans. to the ques. no-03

Huffman code: Given:

Symbol	Probabilities
S_1	0.19
S_2	0.21
S_3	0.25
S_4	0.08
S_5	0.16
S_6	0.06
S_7	0.02
S_8	0.03

Probabilities in descending order:

0.25, 0.21, 0.19, 0.16, 0.08, 0.06, 0.03, 0.02

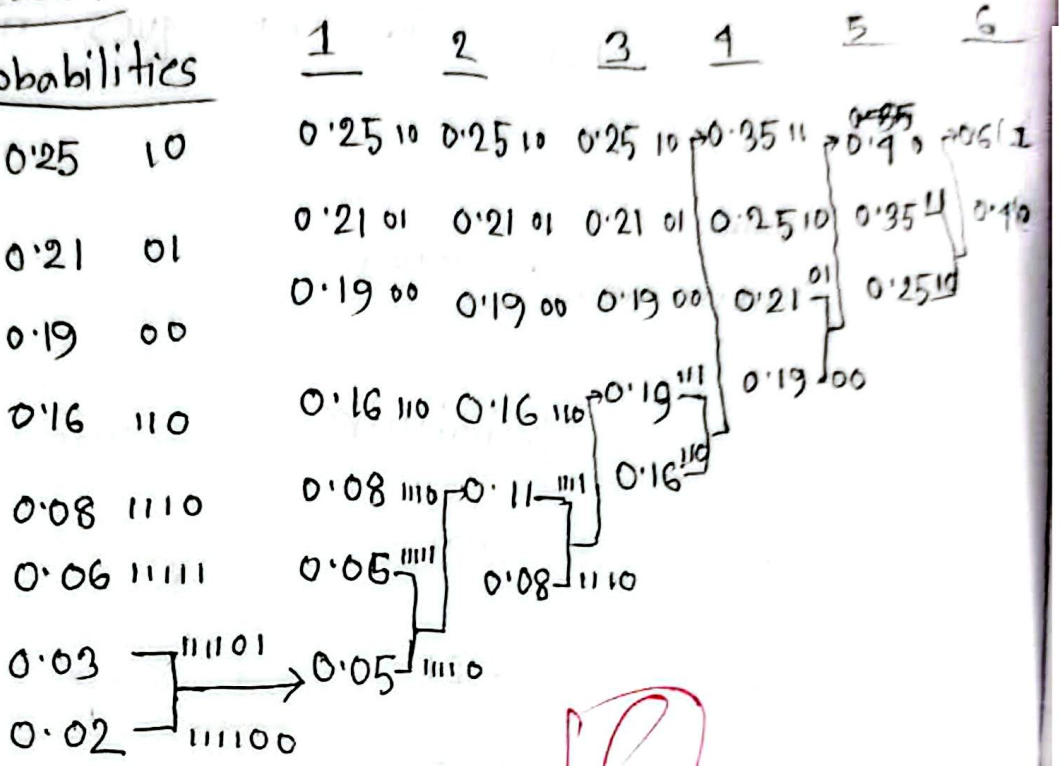
P.T.O

ii) Spatial Redundancy: Neighbouring pixels are highly relatable. Adjacent ~~inten~~ pixels often have same intensity.

iii) Psychovisual Redundancy / Irrelevant information: Some information does not noticed by the human eye. So, removing these details does not affect the image visual quality.

Huffman code:

Symbol	Probabilities	
S0	0.25	10
	0.21	01
	0.19	00
	0.16	110
	0.08	1110
	0.06	11111
	0.03	111101
	0.02	111100


1
2
3
4
5
6
0.25 10
0.21 01
0.19 00
0.16 110
0.08 1110
0.06 11111
0.03
0.02
111101
111100
0.05 11110
0.11 1111
0.08 1110
0.16 110
0.19 111
0.25 10
0.35 11
0.21 01
0.19 00
0.16 110
0.08 1110
0.06 11111

Ans. to the ques. no-02

~~There~~ Redundancy: Redundancy is the extra or repeated information present in an image.
Removing these extra information does not affect the image quality.

There are three types of redundancies present in an image. These are:

i) Coding Redundancy: Occurs when inefficient codes are used.

Example: Using 8 bits when 5 bits are enough.